Trevor Gore's guitar plate thicknessing equations

Share

What are the equations for calculating plate thickness for a guitar top and back and for measure to of a guitar brace according to Trevor gore?

Searched the web >

Good — I have enough to give you a solid overview. Here's a breakdown of Trevor Gore's equations from *Contemporary Acoustic Guitar Design and Build* (with co-author Gerard Gilet):

Trevor Gore's Plate Thicknessing Method

The core idea is to measure the actual material properties of the specific piece of wood you'll use (via tap tones), then calculate the target thickness that will produce consistent vibrational behavior regardless of which piece of wood you're working with.

Step 1: Measure the Elastic Constants via Tap Tones

On a rectangular test plate (the actual piece you're going to use), you tap to find three resonant frequencies while holding the plate at its nodal points:

- **f_L** — longitudinal (long grain) bending mode
- **f_C** — cross grain bending mode
- **f_LC** — torsional (twist) mode

You also measure: length **l**, width **w**, thickness **h**, and mass (to get density **ρ = mass / (l × w × h)**)

Step 2: Calculate Elastic Moduli

Reply...

Sonnet 4.6

Claude is AI and can make mistakes. Please double-check responses.